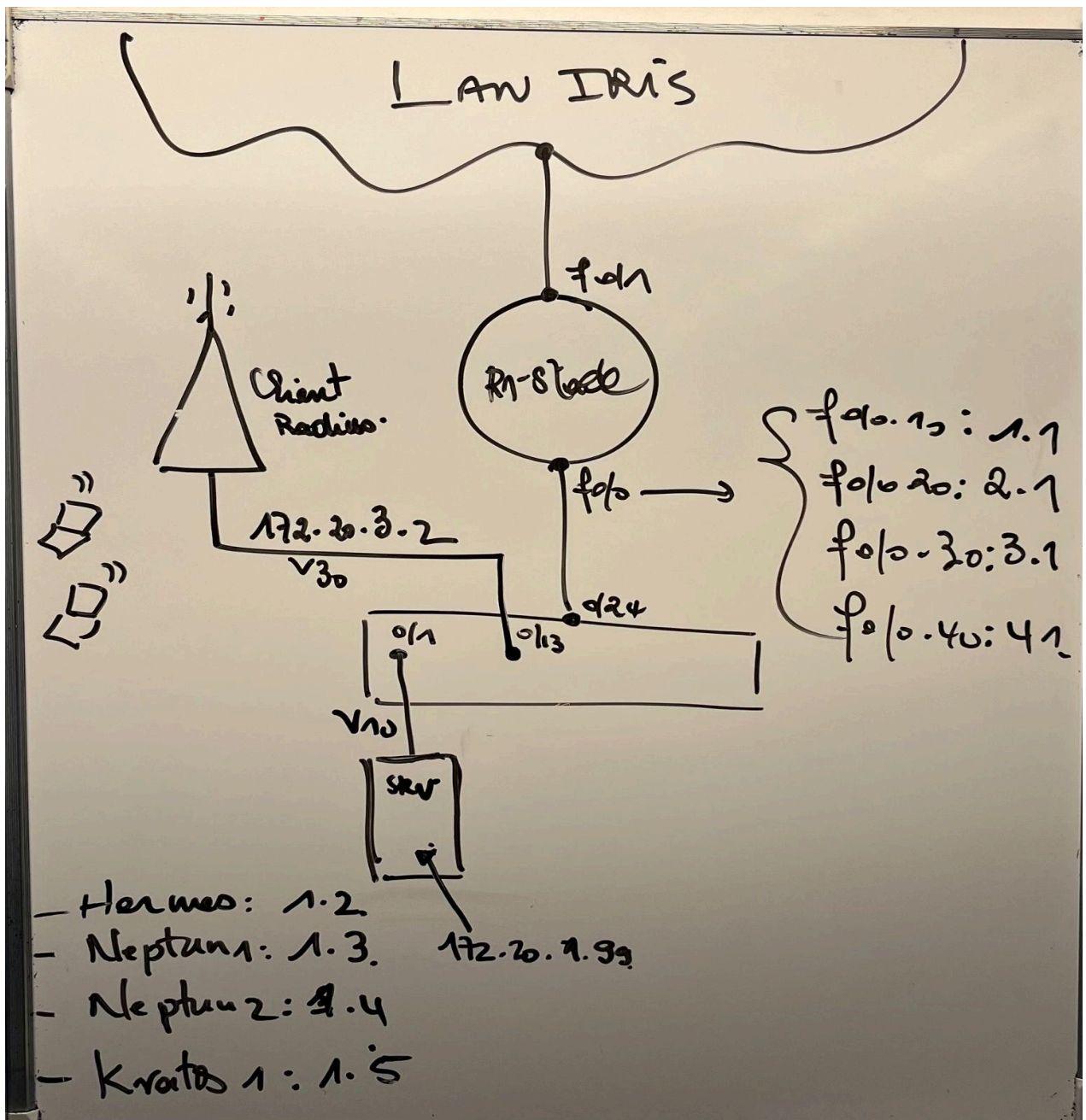
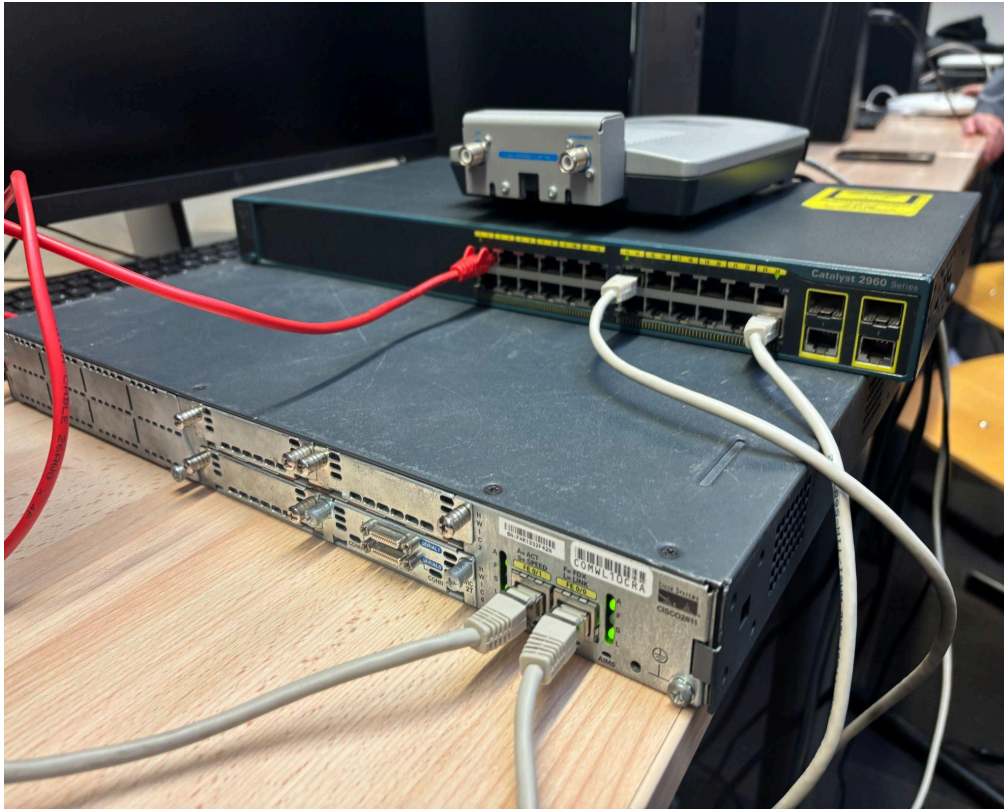


MISSION 6 : MISE EN PLACE D'UNE AUTHENTIFICATION RADIUS





configuration du Routeur :

```
routeur# conf t
```

```
routeur(config)# hostname R1-Stade
```

```
R1-Stade(config)# interface FastEthernet 0/0
```

```
R1-Stade(config-subif)# ip nat inside
```

```
R1-Stade(config-subif)# no shutdown
```

```
R1-Stade(config)# interface FastEthernet 0/1
```

```
R1-Stade(config-subif)# ip nat outside
```

```
R1-Stade(config-subif)# no shutdown
```

```
R1-Stade(config)# interface FastEthernet 0/0.10
```

```
R1-Stade(config-subif)# encapsulation dot1Q 10
```

```
R1-Stade(config-subif)# ip address 172.20.1.1 255.255.255.0
```

```
R1-Stade(config-subif)# ip nat inside
```

```
R1-Stade(config-subif)# no shutdown
```

```
R1-Stade(config)# interface FastEthernet 0/0.20
```

```
R1-Stade(config-subif)# encapsulation dot1Q 20
```

```
R1-Stade(config-subif)# ip address 172.20.2.1 255.255.255.0
```

```
R1-Stade(config-subif)# ip helper-address 172.20.1.5
```

```
R1-Stade(config-subif)# ip nat inside
```

```
R1-Stade(config-subif)# no shutdown
```

```
R1-Stade(config)# interface FastEthernet 0/0.30
```

```
R1-Stade(config-subif)# encapsulation dot1Q 30
```

```
R1-Stade(config-subif)# ip address 172.20.3.1 255.255.255.0
```

```
R1-Stade(config-subif)# ip helper-address 172.20.1.5
```

```
R1-Stade(config-subif)# ip nat inside
```

```
R1-Stade(config-subif)# no shutdown
```

```
R1-Stade(config)# interface FastEthernet 0/0.40
```

```
R1-Stade(config-subif)# encapsulation dot1Q 40
```

```
R1-Stade(config-subif)# ip address 172.20.4.1 255.255.255.0
```

```
R1-Stade(config-subif)# ip nat inside
```

```
R1-Stade(config-subif)# no shutdown
```

```
R1-Stade(config)# ip nat inside source liste 10 interface FastEthernet 0/1 overload
```

```
R1-Stade(config)# ip nat inside source liste 20 interface FastEthernet 0/1 overload
```

```
R1-Stade(config)# ip nat inside source liste 30 interface FastEthernet 0/1 overload
```

```
R1-Stade(config)# ip nat inside source liste 40 interface FastEthernet 0/1 overload
```

```
R1-Stade(config)# access-list 10 permit 172.20.1.0 255.255.255.0
R1-Stade(config)# access-list 20 permit 172.20.2.0 255.255.255.0
R1-Stade(config)# access-list 30 permit 172.20.3.0 255.255.255.0
R1-Stade(config)# access-list 40 permit 172.20.4.0 255.255.255.0
```

```
R1-Stade#show run
Building configuration...

Current configuration : 1935 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R1-Stade
!
boot-start-marker
boot-end-marker
!
logging message-counter syslog
!
no aaa new-model
memory-size iomem 25
!
dot11 syslog
ip source-route
!
!
!
ip cef
!
!
!
multilink bundle-name authenticated
!
!
!
!
!
!
archive
  log config
  hidekeys
!
!
!
!
!
!
!
```



```
interface FastEthernet0/0
  no ip address
  ip nat inside
  ip virtual-reassembly
  duplex auto
  speed auto
!
interface FastEthernet0/0.10
  encapsulation dot1Q 10
  ip address 172.20.1.1 255.255.255.0
  ip nat inside
  ip virtual-reassembly
!
interface FastEthernet0/0.20
  encapsulation dot1Q 20
  ip address 172.20.2.1 255.255.255.0
  ip helper-address 172.20.1.5
  ip nat inside
  ip virtual-reassembly
!
interface FastEthernet0/0.30
  encapsulation dot1Q 30
  ip address 172.20.3.1 255.255.255.0
  ip helper-address 172.20.1.5
  ip nat inside
  ip virtual-reassembly
!
interface FastEthernet0/0.40
  encapsulation dot1Q 40
  ip address 172.20.4.1 255.255.255.0
  ip nat inside
  ip virtual-reassembly
!
interface FastEthernet0/1
  ip address dhcp
  ip nat outside
  ip virtual-reassembly
  duplex auto
  speed auto
!
interface Serial0/0/0
  no ip address
  shutdown
  clock rate 2000000
!
interface Serial0/0/1
```

```
interface Serial0/0/0
  no ip address
  shutdown
  clock rate 2000000
!
interface Serial0/0/1
  no ip address
  shutdown
  clock rate 2000000
!
ip forward-protocol nd
no ip http server
no ip http secure-server
!
!
ip nat inside source list 10 interface FastEthernet0/1 overload
ip nat inside source list 20 interface FastEthernet0/1 overload
ip nat inside source list 30 interface FastEthernet0/1 overload
ip nat inside source list 40 interface FastEthernet0/1 overload
!
access-list 10 permit 172.20.1.0 0.0.0.255
access-list 20 permit 172.20.2.0 0.0.0.255
access-list 30 permit 172.20.3.0 0.0.0.255
access-list 40 permit 172.20.4.0 0.0.0.255
!
!
!
!
!
control-plane
!
!
line con 0
line aux 0
line vty 0 4
  login
!
scheduler allocate 20000 1000
end

R1-Stade#
```

configuration du Switch :

```
switch# conf t
switch(config) #vlan 10
switch(config) #name VLAN10
switch(config) #vlan 20
switch(config) #name VLAN20
switch(config) #vlan 30
switch(config) #name VLAN30
switch(config) #vlan 40
switch(config) #name VLAN40
```

```
switch# conf t
switch# interface range fastEthernet 0/1
switch# switchport mode access
switch# switch port access vlan 10
switch# no shutdown
switch# exit
```

```
switch# conf t
switch# interface range fastEthernet 0/13
switch# switchport mode access
switch# switch port access vlan 30
switch# no shutdown
switch# exit
```

```
Switch#show run
Building configuration...

Current configuration : 2370 bytes
!
version 12.2
no service pad
service timestamps debug uptime
service timestamps log uptime
no service password-encryption
!
hostname Switch
!
!
no aaa new-model
ip subnet-zero
!
!
!
no file verify auto
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
interface FastEthernet0/1
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/2
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/3
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/4
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/5
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/6
  switchport access vlan 10
```



```
!  
interface FastEthernet0/7  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/8  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/9  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/10  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/11  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/12  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/13  
  switchport access vlan 30  
  switchport mode access  
!  
interface FastEthernet0/14  
  switchport access vlan 30  
  switchport mode access  
!  
interface FastEthernet0/15  
  switchport access vlan 30  
  switchport mode access  
!  
interface FastEthernet0/16  
  switchport access vlan 30  
  switchport mode access  
!  
interface FastEthernet0/17  
  switchport access vlan 30  
  switchport mode access  
!  
interface FastEthernet0/18  
  switchport access vlan 30  
  switchport mode access
```

```
    switchport mode access
    !
interface FastEthernet0/18
    switchport access vlan 30
    switchport mode access
    !
interface FastEthernet0/19
    switchport access vlan 30
    switchport mode access
    !
interface FastEthernet0/20
    switchport access vlan 30
    switchport mode access
    !
interface FastEthernet0/21
    switchport access vlan 30
    switchport mode access
    !
interface FastEthernet0/22
    switchport access vlan 30
    switchport mode access
    !
interface FastEthernet0/23
    switchport access vlan 30
    switchport mode access
    !
interface FastEthernet0/24
    switchport mode trunk
    !
interface GigabitEthernet0/1
    !
interface GigabitEthernet0/2
    !
interface Vlan1
    no ip address
    no ip route-cache
    !
ip http server
    !
control-plane
    !
    !
line con 0
line vty 5 15
    !
end

Switch#
```

configuration borne WIFI :

```
ap# conf t
ap(config)# interface BVI 1
ap(config-if)# ip address 172.20.3.2 255.255.255.0
ap(config-if)#no shutdown
ap(config-if)#exit
ap(config)#interface dot11Radio 0
ap(config-if)#no shutdown
```

```
ap>en
Password:
ap#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ap(config)#interface BVI 1
ap(config-if)#ip address 172.20.3.2 255.255.255.0
ap(config-if)#no shu
ap(config-if)#no shutdown
ap(config-if)#exit
ap(config)#interface dot11Radio 0
ap(config-if)#no shutdown
ap(config-if)#
*Mar 10 23:54:00.522: %LINK-5-CHANGED: Interface Dot11Radio0, changed state to reset
*Mar 10 23:54:00.523: %DOT11-4-NO_SSID_OR_NO_VLAN: No SSID configured. Dot11Radio0 not started.
```

```
ap#ping 172.20.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.20.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/201/1002 ms
ap#
```

Document sans titre - Google Doc

Cisco IOS Series AP - Home

Non sécurisé 172.20.3.2

Conversation

CISCO

HOME

EXPRESS SET-UP

EXPRESS SECURITY

NETWORK MAP

ASSOCIATION

NETWORK INTERFACES

SECURITY

SERVICES

WIRELESS SERVICES

SYSTEM SOFTWARE

EVENT LOG

Cisco Aironet 1200 Series Access Point

ap uptime is 3 hours, 31 minutes

Hostname ap

Home: Summary Status

Association

Clients 1Infrastructure clients: 0

Network Identity

IP Address172.20.3.2

MAC Address0013.6072.ae61

Network Interfaces

Interface	MAC Address	Transmission Rate
FastEthernet	0013.6072.ae61	100Mb/s
Radio0-802.11G	0013.1a90.e470	54.0Mb/s
Radio1-802.11A	0013.6057.ea50	54.0Mb/s

Event Log

Time	Severity	Description
Mar 11 03:19:04.999	Warning	RADIUS server 172.20.1.2:1645,1646 has returned.
Mar 11 03:19:04.999	Warning	RADIUS server 172.20.1.2:1645,1646 is not responding.
Mar 11 03:18:52.131	Debugging	Station 6c6a.775f.c986 Authentication failed
Mar 11 03:16:21.693	Debugging	Station 6c6a.775f.c986 Authentication failed
Mar 11 03:16:21.692	Warning	RADIUS server 172.20.1.2:1645,1646 has returned.
Mar 11 03:16:21.692	Warning	RADIUS server 172.20.1.2:1645,1646 is not responding.

Document sans titre - Google Doc X Cisco IOS Series AP - Express Secu X

Non sécurisé 172.20.3.2/ap_express-security.shtml Résumer

ASSOCIATION
NETWORK
INTERFACES
SECURITY
SERVICES
WIRELESS SERVICES
SYSTEM SOFTWARE
EVENT LOG

SSID Configuration

1. SSID ☐ Broadcast SSID in Beacon

2. VLAN
☒ No VLAN ☐ Enable VLAN ID: (1-4094) ☐ Native VLAN

3. Security
☐ No Security
☒ Static WEP Key
☐ EAP Authentication
☐ WPA

Key 1 128 bit

RADIUS Server: 172.20.1.2 (Hostname or IP Address)
RADIUS Server Secret:

RADIUS Server: 172.20.1.2 (Hostname or IP Address)
RADIUS Server Secret:

Apply Cancel

SSID Table

Delete	SSID	VLAN	Encryption	Authentication	Key Management	Native VLAN	Broadcast SSID
☒	CLIENT-RADIUS-NYT	none	wep mandatory	open+EAP , network EAP	none		✓

 MSG_IRIS-Paris_Etudiants

 CLIENT-RADIUS-NYT
Sécurisé

Entrez votre nom d'utilisateur et votre mot de passe



CLIENT-RADIUS-NYT

Sécurisé

Poursuivre la connexion ?

Si vous vous attendez à trouver CLIENT-RADIUS-NYT à cet emplacement, connectez-vous. Dans le cas contraire, il s'agit d'un autre réseau portant le même nom.

Afficher les détails du certificat

Se connecter

Annuler



Wi-Fi



CLIENT-RADIUS-NYT

Connecté, sécurisé



Déconnecter

Carte réseau sans fil Wi-Fi :

Suffixe DNS propre à la connexion. . . :
Adresse IPv6 de liaison locale. . . . : fe80::ae08:bc2f:d625:ae4a%16
Adresse IPv4. : 172.20.3.100
Masque de sous-réseau. : 255.255.255.0
Passerelle par défaut. : 172.20.3.1

Carte Ethernet Connexion réseau Bluetooth :

Statut du média. : Média déconnecté
Suffixe DNS propre à la connexion. . . :

PS C:\Users\zidan> ping 1.1.1.1

Envoi d'une requête 'Ping' 1.1.1.1 avec 32 octets de données :

Réponse de 1.1.1.1 : octets=32 temps=5 ms TTL=55

Réponse de 1.1.1.1 : octets=32 temps=8 ms TTL=55

Réponse de 1.1.1.1 : octets=32 temps=7 ms TTL=55

Réponse de 1.1.1.1 : octets=32 temps=5 ms TTL=55

Statistiques Ping pour 1.1.1.1:

Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),

Durée approximative des boucles en millisecondes :

Minimum = 5ms, Maximum = 8ms, Moyenne = 6ms

PS C:\Users\zidan>